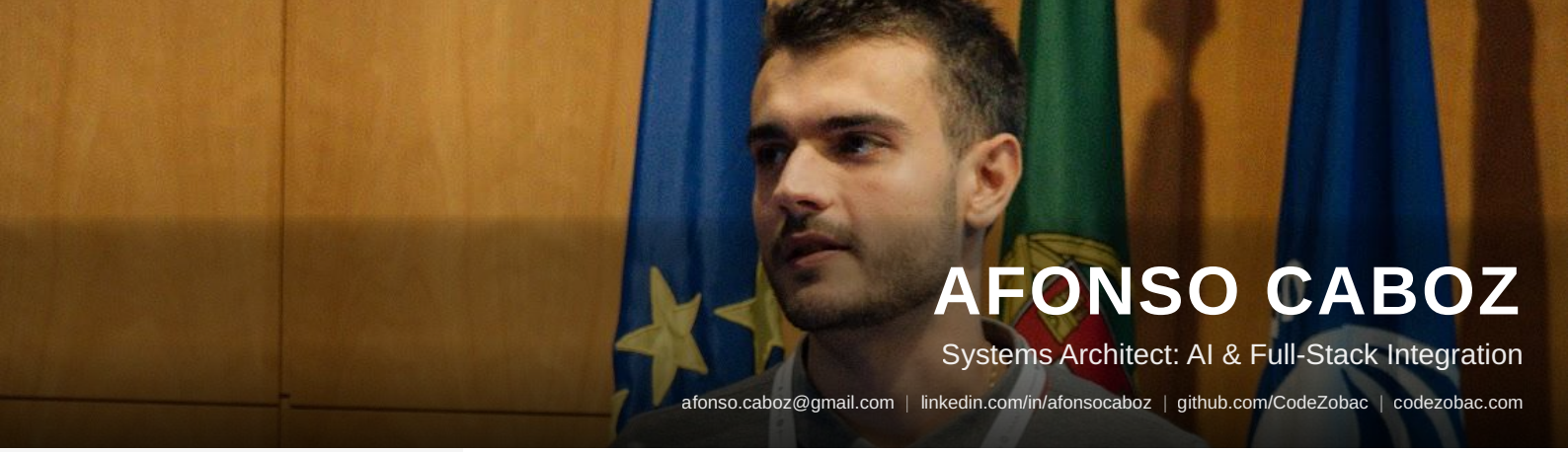


AFONSO CABOZ

Systems Architect: AI & Full-Stack Integration

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PHILOSOPHY

"Code is a liability.
A solution is an asset."

My guiding principle is to write the least amount of code necessary to create the most value.

TECH STACK

AI / LLM

RAG Pipelines | CrewAI | Langfuse | LiteLLM | Observability

Backend & Systems

Python | FastAPI | Django | Rust | Axum | Tauri | PostgreSQL

Frontend

Next.js | Vite | TypeScript | Tailwind CSS

Infrastructure

Docker | Terraform | CI/CD

EDUCATION

Teesside University

Bachelor in Software Engineering

Top up year

2026 to 2027 • Ongoing

etic_ ESCOLA DE TECNOLOGIAS INOVAÇÃO E CRIAÇÃO DO ALGARVE

Web development technical degree

2025 • Final grade: 18/20

I help teams make AI applications reliable, traceable, and economical to operate. My work addresses unreliable agent outputs, rising API costs, and software that has become difficult to maintain.

In **FleetFlow,** I combined Langfuse and LiteLLM observability with automated testing and LLM as a Judge evaluations to assess response quality, investigate failures, and control costs. At **ETIC Algarve**, I audited a struggling legacy system and delivered a rebuilt MVP, starting with a working prototype within 48 hours.

I make technical complexity easier to navigate. By mapping system dependencies and data flows, I identify potential failure points and compare architectural approaches before implementation. This gives teams clearer proposals and a concrete basis for technical decisions.

I ground AI compliance in observable evidence. Using Langfuse, I trace prompts, context, tool interactions, and outputs to make agent behaviour inspectable. Combined with systematic evaluation, these records support traceability, risk assessment, and human oversight.

I work with **Python, Next.js, TypeScript, and Rust**, connecting AI orchestration, backend services, and user interfaces into applications teams can operate and maintain.

My personal development centres on awareness, reflection, and deliberate practice. Spiritual growth is part of that process: I practise Reiki and use Ayurvedic principles to guide my dietary choices. These practices give me space to examine my habits and bring greater intention to how I live, learn, and relate to others.

Selected Projects

CLIENT PROJECTS

ETIC_Algarve Resource Hub: Unified Institutional Management System

My role. Solutions Architect & Lead Engineer

Engineered a high-performance ecosystem to unify fragmented institutional resources. The architecture features advanced search with intelligent autocomplete and secure identity management via **Supabase/ Google OAuth (RBAC)**. To ensure operational integrity, I implemented automated communication via **Resend API** and validated system stability through **k6 load testing** (50+ concurrent users). Deployment is fully automated using **Terraform**, **Docker Compose**, and a **Makefile**-based CI/CD pipeline with **Nginx SSL** termination.

SKILLS AND DELIVERABLES

CI/CD | PERFORMANCE TESTING | LOAD TESTING | TERRAFORM | DOCKER

FleetFlow: AI-Powered Autonomous Fleet Management Platform

My role. AI Solutions Architect: LLMOps & Agentic Systems

Engineered the transition of FleetFlow from simple task-driven assistants to fully autonomous, **RAG-powered agents**. Implemented a multi-layer observability architecture using **Langfuse** and **LiteLLM**, utilizing an "**LLM-as-a-Judge**" framework to reduce API costs while increasing output precision. Developed a predictive "**Fuel Intelligence Layer**" for anomaly and fraud detection via advanced algorithmic modeling. Additionally, engineered an **AI Automated Testing** pipeline to mitigate model hallucinations, ensuring a self-correcting, reliable production environment.

SKILLS AND DELIVERABLES

MLOPS | LANGFUSE | AI AGENT DEVELOPMENT | PREDICTIVE ANALYTICS | AI IMPLEMENTATION

SOFT SKILLS

Forged in high-pressure hospitality environments and refined through complex technical work, these strengths shape how I collaborate, decide, and support teams.

Empathetic Thinking

Out-of-the-box Perspective

Mental Visualization

Associative Memory

Present Attitude

Nurturing Environments

PERSONAL PROJECTS

CyberCompass: Agentic Cyber-Ethics Learning Platform

My role. Full-Stack AI Solutions Architect

Engineered a bilingual cyber-ethics platform delivering adaptive training across disinformation, deepfake detection, cyberbullying, and online identity safety. Architected a three-service system with a **Next.js 15/React 19** frontend and dedicated **Rust/Axum** database and AI orchestration services. Implemented real-time agentic simulations over **WebSockets** with pluggable LLM providers, structured prompting, safety classification, deterministic scoring and fallbacks, plus multi-layer **Langfuse/ OpenTelemetry** observability for quality, cost, and risk attribution.

SKILLS AND DELIVERABLES

RUST | AGENTIC AI & LLM ORCHESTRATION | PROMPT ENGINEERING & AI SAFETY | WEBSOCKETS & REAL-TIME | LANGFUSE OBSERVABILITY

IN Sintonia: Agentic Ayurvedic Nutrition Platform

My role. Full-Stack AI Solutions Architect

Engineered an agentic real-time platform to align dietary interventions with Ayurvedic principles. Implemented a highly efficient **Mixture-of-Experts (MoE)** architecture using hierarchical prompts and a "**root-level**" verification layer to eliminate semantic drift. Architected a high-performance **Rust** backend leveraging **ZeroClaw** for AI orchestration, utilizing **Axum** and **Utopia** for high-throughput APIs. Integrated **WebSockets** to facilitate seamless, real-time nutritional guidance and continuous data streams via a modern **Next.js/Tailwind** frontend.

SKILLS AND DELIVERABLES

AI MODEL INTEGRATION | AI AGENT DEVELOPMENT | RUST | LANGCHAIN | AI COMPLIANCE

Employment History

AI Solutions Architect | LLMOps & Agentic Systems

Feb 2026 – May 2026

VivaDrive

Project Overview: Scaling the FleetFlow AI-powered fleet management platform from reactive assistants to fully autonomous, RAG-driven agents.

Key Interventions:

- **Unified Observability & Cost Optimization:** Engineered a multi-layer observability architecture using Langfuse and LiteLLM. Implemented an "LLM-as-a-Judge" framework, reducing API costs via gpt-4o-mini while increasing output precision through total request traceability.
- **Predictive Intelligence Development:** Developed the core "Fuel Intelligence Layer," transitioning architecture from static data processing to a predictive Data Warehouse model for automated anomaly and fraud detection.
- **Autonomous Reliability:** Engineered an AI Automated Testing pipeline to mitigate model hallucinations, creating a self-correcting ecosystem where specialized models validate autonomous agent outputs.

Result: Transformed FleetFlow into a resilient, predictive architecture capable of managing complex logistics with minimal human intervention.

Solutions Architect | Systemic Debt Recovery & MVP Deployment

Aug 2025 – Sep 2025

ETIC_Algarve

Project Overview: Rehabilitating a legacy resource management system at ETIC Algarve burdened by mounting technical debt and architectural entropy.

Key Interventions:

- **Architectural Audit:** Conducted deep-dive audits to identify root causes of systemic failure, addressing fundamental structural decay rather than superficial UI fixes.
- **Strategic Rebuild Advocacy:** Engineered a framework to pivot from high-maintenance refactoring toward a modern, scalable architecture for long-term maintainability.
- **Rapid Prototyping (48-Hour Execution):** Developed a functional, high-fidelity prototype within 48 hours to validate technical viability and secure stakeholder buy-in.
- **Stakeholder Synchronization:** Aligned complex engineering milestones with organizational objectives and institutional feedback loops.

Result: Delivered a production-ready MVP, transforming a failing legacy asset into a scalable, high-performance platform.

Seeking to connect with a visionary team where I can contribute to and learn from a world-class engineering culture.